*International Journal of Sports Physiology and Performance*, (Ahead of Print)
https://doi.org/10.1123/ijspp.2019-0345
© 2020 Human Kinetics, Inc.

Human Kinetics
ORIGINAL INVESTIGATION

# Topical Sodium Bicarbonate: No Improvement in Blood Buffering Capacity or Exercise Performance

**Alannah K.A. McKay, Peter Peeling, Martyn J. Binnie, Paul S.R. Goods, Marc Sim, Rebecca Cross, and Jason Siegler**

***Purpose:*** To assess the efficacy of a topical sodium bicarbonate (0.3 g/kg body weight $NaHCO_3$) application (PR lotion; Amp Human) on blood buffering capacity and performance in recreationally active participants (study A) and moderately trained athletes (study B). ***Methods:*** In Study A, 10 participants completed 2 experimental trials: oral $NaHCO_3$ (0.3 g/kg body weight + placebo lotion) or PR lotion (0.9036 g/kg body weight + oral placebo) applied 90 minutes prior to a cycling task to exhaustion (30-s sprints at 120% peak power output with 30-s rest). Capillary blood was collected and analyzed for pH, bicarbonate, and lactate every 10 minutes throughout the 90-minute loading period and postexercise at 5, 10, and 15 minutes. In Study B, 10 cyclists/triathletes completed 2 experimental trials, applying either PR or placebo lotion 30 minutes prior to a cycling performance task (3 × 30-s maximal sprints with 90-s recovery). Capillary blood samples were collected at baseline, preexercise, and postexercise and analyzed as per study A. ***Results:*** In Study A, pH and bicarbonate were significantly elevated from baseline after 10 minutes in the oral $NaHCO_3$ condition and throughout recovery compared with no elevation in the PR lotion condition ($P < .001$). No differences in cycling time occurred between PR lotion (349 [119] s) and oral $NaHCO_3$ (363 [80] s; $P = .697$). In Study B, no differences in blood parameters, mean power ($P = .108$), or peak power ($P = .448$) were observed between conditions. ***Conclusions:*** PR lotion was ineffective in altering blood buffering capacity or enhancing performance in either trained or untrained individuals.

***Keywords:*** pH, athletes, $HCO_3^-$

The ingestion of sodium bicarbonate ($NaHCO_3$) is well accepted as an efficacious ergogenic aid to improve short-duration, high-intensity exercise performance.[1] The exogenous intake of $NaHCO_3$ acts as an extracellular buffer, raising blood pH and bicarbonate ($HCO_3^-$) concentrations and augmenting the rate of hydrogen ion ($H^+$) efflux from the muscle.[2,3] Meta-analyses have demonstrated improvements of ~2% to 3% across a range of performance measures after $NaHCO_3$ ingestion, with larger effects evident when repeated bouts of high-intensity exercise with incomplete recovery are performed.[4,5] However, some contradictory performance outcomes are also reported, likely due to variations in supplementation protocols (eg, disparities in dose, timing, and ingestion methods) or interindividual responses to the supplement and the ensuing performance task.[4,6,7]

Nevertheless, contemporary recommendations suggest the optimal administration protocol for oral $NaHCO_3$ ingestion to be between 0.2 and 0.4 g/kg body weight (BW) and consumed 60 to 150 minutes prior to exercise.[4,7] Larger doses (>0.4 g/kg BW) may induce a greater degree of metabolic alkalosis,[4,8] but are generally not well tolerated by the gastrointestinal (GI) tract. Indeed, GI distress is a common side effect of $NaHCO_3$ supplementation, with nausea, stomach pain, diarrhea, and vomiting all frequently reported symptoms.[9,10] Once ingested, $NaHCO_3$ dissociates in the stomach to $Na^+$ and $HCO_3^-$. Although the relatively large amount of $HCO_3^-$ is quickly neutralized by $H^+$ in the stomach, this process also generates greater carbon dioxide tension, which, if excessive, can induce GI discomfort.[11] Furthermore, the large amount of $Na^+$ ingested can irritate the intestinal mucosa and prompt osmotic fluctuations resulting in diarrhea.[12] Unsurprisingly, the presence of significant GI distress has been suggested to mitigate the performance-enhancing potential of $NaHCO_3$ supplementation.[13] Although practical strategies to minimize GI discomfort have emerged, including serial loading protocols[14,15] and the coingestion of $NaHCO_3$ with a carbohydrate-rich meal,[16] the potential for experiencing negative side effects remains a key consideration for athletes using this ergogenic aid. Athletes potentially benefitting from supplementation may ultimately be discouraged from use due to fear of GI distress, particularly around competition.

With this in mind, a method of $NaHCO_3$ administration that bypasses the GI tract may be a favorable alternative to oral supplementation. One such prospect is a newly developed topical $NaHCO_3$ cream (PR lotion; Amp Human Performance, Park City, UT), which proposes to deliver $NaHCO_3$ locally across the skin through a transdermal delivery system. According to the manufacturer, PR lotion is designed to fluidize the outer layer of the skin and modulate the tight junctions in the epidermis, creating a delivery pathway by which $NaHCO_3^-$ can reach the bloodstream. Presently, no peer-reviewed literature on this product exists, and the extent to which this application method compares to the increase in blood buffering capacity and ~2% to 3% performance improvement consistently observed after oral ingestion of $NaHCO_3$ is unknown. As such, the aim of this investigation was 2-fold. The initial aim

McKay, Peeling, Binnie, and Goods are with the School of Human Sciences (Exercise and Sport Science), University of Western Australia, Crawley, WA, Australia, and the Western Australian Inst of Sport, Mt Claremont, WA, Australia. McKay is also with the Australian Inst of Sport, Bruce, ACT, Australia. Sim is with the School of Medical and Health Sciences, Edith Cowan University, Joondalup, WA, Australia, and the Medical School, Royal Perth Hospital Unit, University of Western Australia, Perth, WA, Australia. Cross and Siegler are with the School of Science and Health, Sport and Exercise Science, Western Sydney University, Penrith, NSW, Australia. McKay (alannah.mckay@ausport.gov.au) is corresponding author.


Authenticated t_lambrinides@yahoo.com | Downloaded 06/23/20 08:35 PM UTC

(study A) was to profile any changes in blood acid–base balance (pH, $HCO_3^-$, and blood lactate concentrations) after application of PR lotion and to compare those changes with a standard oral load of $NaHCO_3$ (0.3 g/kg BW) at rest and following repeated bouts of high-intensity cycling to failure in recreationally active individuals. Subsequently, study B examined the performance effect of PR lotion on mean and peak power output (PPO) during a predetermined series of repeated Wingate performance tasks in moderately trained athletes.

# Methods

## Participants

This study was conducted independently across 2 research laboratories, with 10 recreationally active men and women completing study A at Western Sydney University (School of Health Sciences, Sport and Exercise Science) and 10 moderately trained male cyclists and triathletes completing study B at the University of Western Australia (School of Human Sciences, Sport and Exercise Science) (see Table 1). Prior to participation, requirements and risks were explained to participants and written informed consent obtained. Ethics approval was granted by the institutional ethics committees at both institutions, respectively. Schematic representations of Studies A and B can be found in Figure 1.

## Study A

Participants attended the climate controlled laboratory (23°C [1°C]) on 3 occasions at the same time of day, with trials separated by a minimum of 5 days but no more than 7 days. On the first visit, participants performed a graded exercise test on a cycle ergometer (Velotron, Seattle, WA) to determine individual workloads for the subsequent experimental trials (120% PPO as determined by Currell et al[17]). Thirty minutes following the graded exercise test and after a standardized warm-up, participants were familiarized with the experimental exercise task and completed 30-second efforts (120% PPO separated by 30-s rest) on the cycle ergometer until volitional exhaustion. This protocol was selected on the basis that repeated sprint exercise of between 1- and 7-minute duration has consistently demonstrated a performance benefit after oral $NaHCO_3$ supplementation.[4,5]

Prior to the first experimental trial, participants completed a 24-hour recall of their dietary intake and were instructed to replicate this dietary pattern (macronutrient composition, volume, and timing) prior to the second and final experimental trial. Upon arriving to the laboratory during both experimental trials and after 10-minute seated rest, duplicate capillary bloods samples were collected from the fingertip into 100-μL glass heparinized capillary tubes and immediately analyzed for baseline blood pH and $HCO_3$ concentrations on an ABL 80 flex series (Radiometer, Copenhagen, Denmark). Thereafter, in a doubled-blind (randomized and counterbalanced) repeated-measures design, participants were then assigned to one of the following 2 conditions:

- Oral $NaHCO_3$ ingestion (0.3 g/kg BW) consumed in gelatin capsules + placebo-matched lotion (Amp Human) (0.9036 g BW; applied 100 min prior to exercise).
- Oral placebo (0.3 g/kg BW) of calcium carbonate consumed in gelatin capsules + PR lotion (Amp Human) (0.9036 g × BW; applied 100 min prior to exercise).

Participants consumed the $NaHCO_3$ or placebo capsules, split over 3 equal doses and consumed within a 30-minute period (30, 15, and 0 min prior to lotion application; see Figure 1). Each $NaHCO_3$ dose was consumed with a standardized snack, providing 10 mL/kg BW fluid containing 1.5 g/kg BW maltodextrin (Poly-Joule; Nutricia, Sydney, Australia). During the final 10 minutes of the 30-minute ingestion period, participants applied either PR or placebo-matched lotion to as much of the lower limb as possible. The amount of lotion (0.9036 g  BW) was determined by accounting for the percentage of $NaHCO_3$ (33.2%) per gram of PR lotion (as stated by the manufacturer) and matched to the oral $NaHCO_3$ load of that individual. The placebo-matched lotion was provided by the manufacturer with $NaHCO_3$ removed but still containing 0.5% menthol as per commercially available PR lotion (https://amphumanperformance.com). Once the ingestion and lotion application protocol was complete, participants remained in a rested state in the laboratory for a further 90 minutes. Throughout this 90-minute period, capillary blood was sampled in duplicate at 10-minute intervals to monitor any changes in acid–base balance. Participant GI discomfort was documented immediately following the sampling period using a 100-mm visual analog scale for 8 different GI symptoms (nausea, flatulence, bloating, belching, stomach ache, bowel urgency, diarrhea, and vomiting).

Participants then completed a standardized warm-up consisting of a 10-minute series of short, high-intensity bouts on the cycle ergometer. This included an initial 5-minute period at 50% PPO followed by 5 minutes of step-wise increment (eg, 60%, 70%, 80% PPO) in resistance for 15-second bursts followed by a return to 50% PPO for 45 seconds. The experimental trial then required participants to perform repeated 30-second efforts at 120% PPO separated by 30-second rest on a cycle ergometer until volitional exhaustion. The exercise task was terminated when participants could no longer maintain a cadence above 60 rpm for 3 or more pedal strokes. The number of efforts completed (and seconds completed during the final 30-s bout) was recorded. Upon immediate completion of both experimental trials, postexercise blood samples were collected for both acid–base balance and blood lactate (Lactate Scout; EKF Diagnostics, Barleben, Germany),

**Table 1 Participant Characteristics for Study A and Study B**

| | Study A | Study B |
|---|---|---|
| Participants | Recreationally trained men (n = 7) and women (n = 3) | Moderately trained male cyclists and triathletes (N = 10) |
| Body weight, kg | 75.0 (6.9) | 80.9 (12.6) |
| Height, cm | 175.8 (6.6) | 179.4 (4.4) |
| Peak power output, W | 267 (65) | 355 (31) |
| $VO_2$max, mL/kg/min | — | 63.3 (10.1) |

Abbreviation: VO2max, maximal oxygen uptake. Note: Results are expressed as mean (SD).



Authenticated t_lambrinides@yahoo.com | Downloaded 06/23/20 08:35 PM UTC

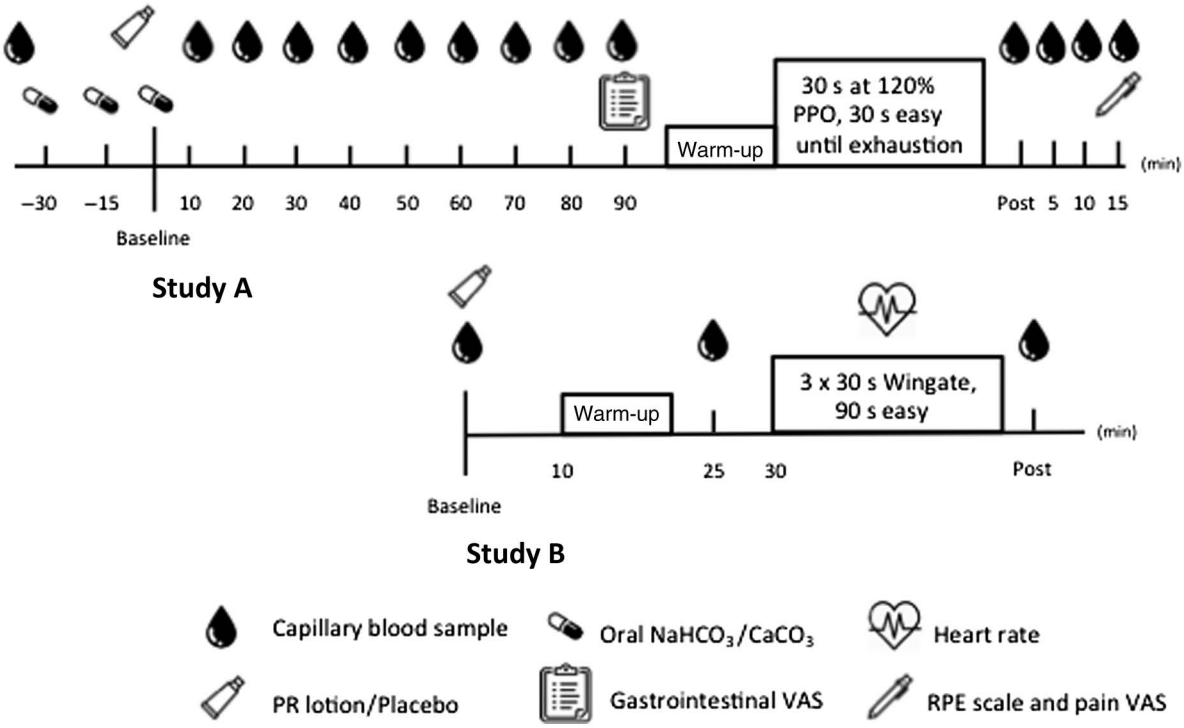


**Figure 1** — Diagrammatic representation of the experimental overview for study A and study B. $CaCO_3$ indicates calcium carbonate; $NaHCO_3$, sodium bicarbonate; PPO, peak power output; RPE, rating of perceived exertion; VAS, visual analog scale.

with additional samples obtained at 5, 10, and 15 minutes postexercise. Subjective perception ratings of pain (100-mm visual analog scale) and fatigue scores (ratings of perceived exertion 6–20 scale[18]) were also obtained at 15 minutes postexercise.

## Study B

In study B, athletes completed 3 testing sessions (one familiarization and 2 experimental trials) at a similar time of day, separated by a minimum of 48 hours. Athletes were instructed to keep training sessions/volume consistent during the 24 hours prior to each experimental trial. During the first session, athletes completed a cycling-based graded exercise test to exhaustion using a stationary air-braked cycle ergometer (Evolution Cycles, Melbourne, Australia) connected to a customized computer program for the determination of power output (Cyclemax; The University Western Australia, Perth Australia). Expired air was analyzed via a metabolic cart (TrueOne 2400; ParvoMedics Inc, Sandy, UT). Following a recovery period, athletes completed a brief familiarization of the experimental trial, which involved a 30-second maximal Wingate cycling test performed on a front access bike, allowing athletes to become accustomed with the exercise protocol and laboratory equipment.

During the next 2 visits, athletes provided a 100-µL capillary blood sample upon arrival to the laboratory. Samples were immediately analyzed for blood lactate, pH, and $HCO_3^-$ using a portable blood gas analyzer (iSTAT; Abbott, Chicago, IL), which was calibrated prior to each session in accordance with manufacturer's guidelines. In a doubled-blind (randomized and counterbalanced) repeated-measures design, athletes then applied 0.9036 g/kg BW of either the PR lotion or the placebo-matched lotion to the entire lower limb (as per study A). During the next trial, athletes were provided with the opposite treatment. Once the lotion was applied, athletes commenced a standardized ~10-minute warm-up (4 min at 50% PPO, 4 min at 65% PPO followed by 3 × 10-s sprints at 150% PPO with 50-s recovery) 15 minutes prior to the start of the experimental protocol. Athletes then rested on the bike for 5 minutes, during which time a second capillary blood sample was collected. Thirty minutes following application of the lotion (which fits within the manufacturer's recommended timing; https://amphumanperformance.com), athletes commenced the experimental protocol. From a standing start, athletes completed 3 × 30-second maximal Wingate sprints on a front access cycle ergometer, separated by 90-second active recovery (<100 W). Peak power, mean power, and peak heart rate (Polar RS400; Polar, Kempele, Finland) were recorded for each sprint. Finally, a capillary blood sample was collected immediately postexercise.

## Statistical Analysis

Results are presented as mean (SD). All statistical analyses were completed using IBM SPSS Statistics (version 25; SPSS Inc, Chicago, IL). All hematological and performance variables were analyzed using a 2-way (condition × time) analysis of variance for repeated measures. Post hoc, least significant differences paired sample *t* tests were used to identify specific trial differences. Significance was set at $P < .05$. When significant differences are stated, Cohen effect sizes (*d*) with 95% confidence intervals (CIs) were calculated.

# Results

## Study A

Blood pH and $HCO_3^-$ profiles for both conditions throughout the 90-minute loading period are summarized in Figure 2. Throughout the loading period, both resting blood pH and $HCO_3^-$ were



Authenticated t_lambrinides@yahoo.com | Downloaded 06/23/20 08:35 PM UTC

significantly elevated from 10 to 90 minutes postingestion in the oral $NaHCO_3$ condition (pH: mean increase of 0.05 a.u. [95% CI, 0.03 to 0.07; $d = 1.31$ (0.61)] from baseline to 90 min; $HCO_3^-$: mean increase of 4.7 mmol/L [95% CI, 3.7 to 5.7; $d = 1.82$ (0.41)] from baseline to 90 min) compared with the PR lotion ($P < .001$; Figure 2). Decreases in pH (oral $NaHCO_3$: −0.17 a.u. [95% CI, −0.15 to −0.20]; PR lotion: −0.18 a.u. [95% CI, −0.14 to −0.22] and $HCO_3^-$ (oral $NaHCO_3$: −14.2 mmol/L [95% CI, −12.3 to −16.2]; PR lotion: −10.9 mmol/L [95% CI, −9.3 to −12.5]) were evident immediately postexercise ($P < .001$), which had started to recover by 10 minutes postexercise ($P = .004$ and $P = .007$, respectively). However, pH and $HCO_3^-$ values were highest during all time points postexercise in the oral $NaHCO_3$ condition, compared with the PR lotion condition ($P < .01$). Blood lactate increased immediately postexercise ($P < .001$), but to a higher end point in the oral $NaHCO_3$ (16.4 [5.0] mmol/L) compared with PR lotion (13.8 [3.8] mmol/L) condition ($d = 0.57$ [0.48], $P = .023$). There were no differences in blood lactate between conditions at 5, 10, or 15 minutes postexercise ($P > .111$).

Total cycling time to volitional exhaustion was 363 (80) seconds in the oral $NaHCO_3$ condition and 349 (119) seconds in the PR lotion condition. No differences in cycling time were evident between conditions ($P = .697$; see Figure 3). No differences in pain scores (oral $NaHCO_3$: 56.4 [24.5]; PR lotion: 49.4 [26.4]; $P = .169$) or GI symptoms ($P = .423$; Figure 4) were observed following supplementation. However, ratings of perceived exertion in the PR Lotion trial were significantly higher than the oral $NaHCO_3$ compared with PR lotion condition (oral $NaHCO_3$: 18.3 [1.0]; PR lotion: 18.9 [0.7]; $d = 0.65$ [0.58], $P = .015$).

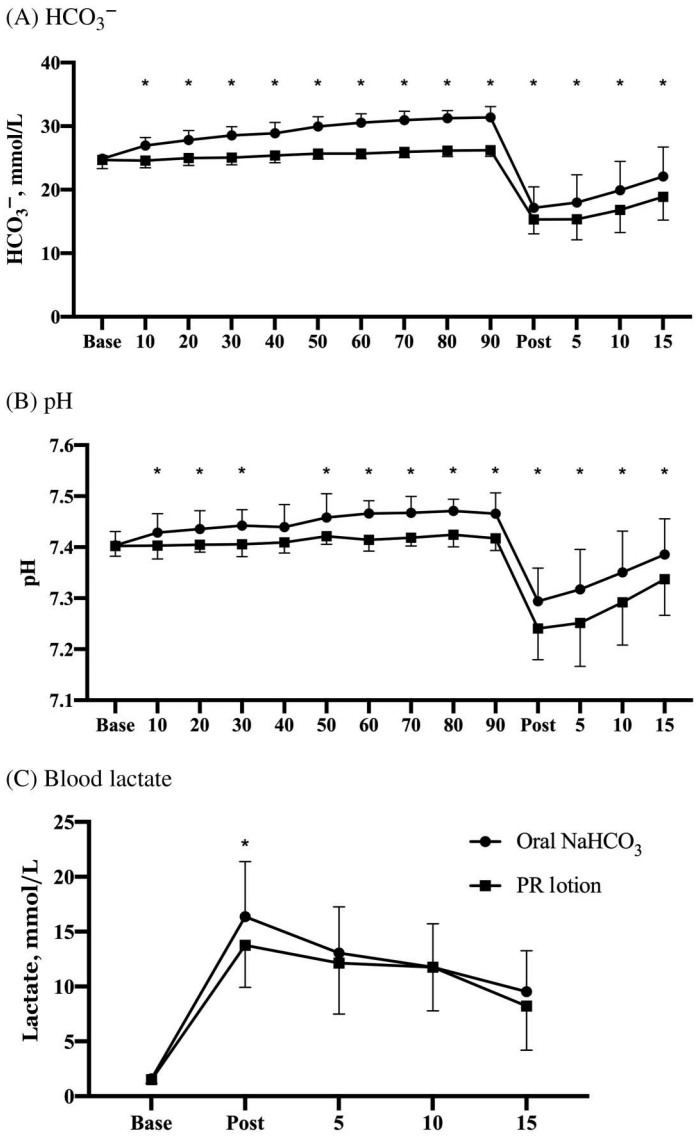


**Figure 2** — (A) Blood $HCO_3^-$, (B) pH, and (C) blood lactate concentrations preexercise and postexercise for the oral $NaHCO_3$ and PR lotion conditions during study A. $HCO_3^-$ indicates bicarbonate; $NaHCO_3$, sodium bicarbonate. *Significant difference from oral $NaHCO_3$ ($P < .05$).

## Study B

Performance variables are shown in Figure 5. During both trials, a decrease in mean power was evident during the Wingate 2 (13.7% decrease, $P = .001$) and Wingate 3 (19.8% decrease; $P = .001$) tests when compared with the Wingate 1 test. A similar outcome was evident for peak power, with a decrease in peak power occurring during Wingate 2 (10.1% decrease; $P = .005$) and Wingate 3 (14.4% decrease; $P = .014$) compared with Wingate 1. Heart rate increased from Wingate 1 (PR lotion: 169 [6] and placebo: 171 [7]) to Wingate 2 (PR lotion: 173 [6] and placebo: 173 [5]; $P < .001$) and further increased by Wingate 3 (PR lotion: 176 [7] and placebo: 174 [6]; $P = .007$). No differences between the PR lotion and placebo conditions were evident for mean power ($P = .108$), peak power ($P = .448$), or heart rate ($P = .971$). Blood $HCO_3^-$ concentrations ($P = .004$) and pH ($P = .021$) decreased from baseline to preexercise, and further decreased postexercise ($P < .001$) (Table 2). No trial differences were evident. Blood lactate increased from baseline to preexercise in the PR lotion condition only ($d = 1.09$ [0.78], $P = .046$), with both the PR lotion and placebo conditions significantly increasing to similar postexercise levels ($P < .001$).

# Discussion

In this study, PR lotion had no influence on blood parameters or performance outcomes in either moderately trained athletes or recreationally active individuals. Of note, increases in blood pH and $HCO_3^-$ concentrations were only evident following oral $NaHCO_3$ supplementation, despite a similar absolute amount of $NaHCO_3$ administered between both oral and topical delivery

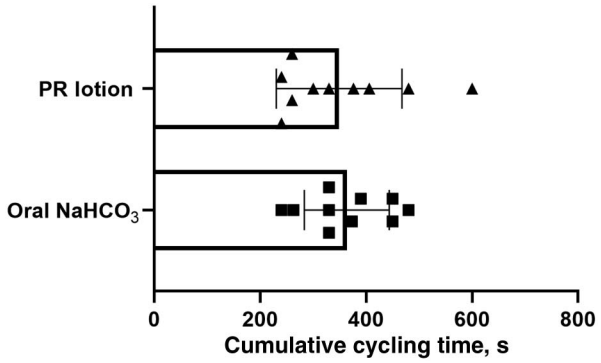


**Figure 3** — Total cycling time to exhaustion (in seconds) for the oral $NaHCO_3$ and PR lotion conditions during study A. $NaHCO_3$ indicates sodium bicarbonate.


Authenticated t_lambrinides@yahoo.com | Downloaded 06/23/20 08:35 PM UTC

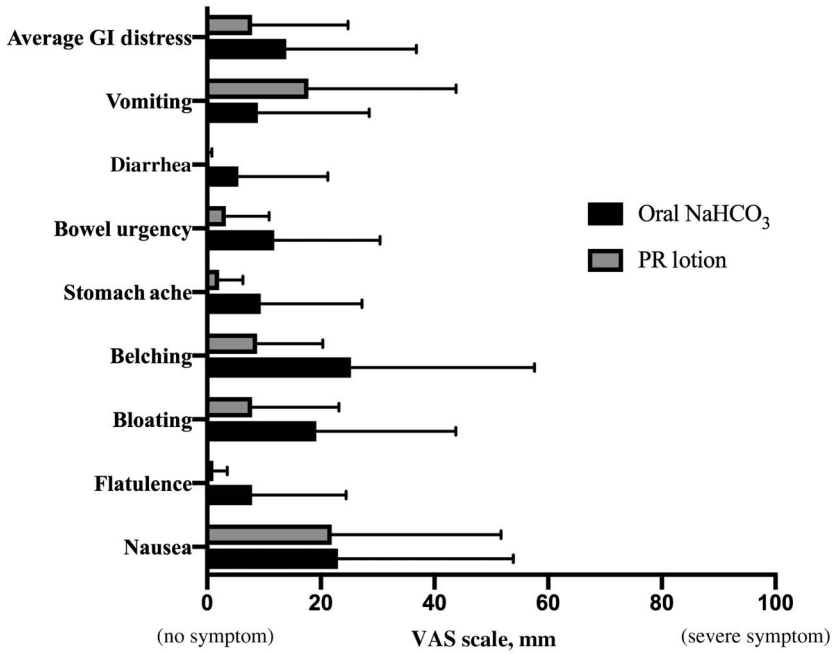


**Figure 4** — Severity of GI symptoms following the supplementation period for the oral $NaHCO_3$ and PR lotion conditions in study A. GI indicates gastrointestinal; $NaHCO_3$, sodium bicarbonate; VAS, visual analog scale.

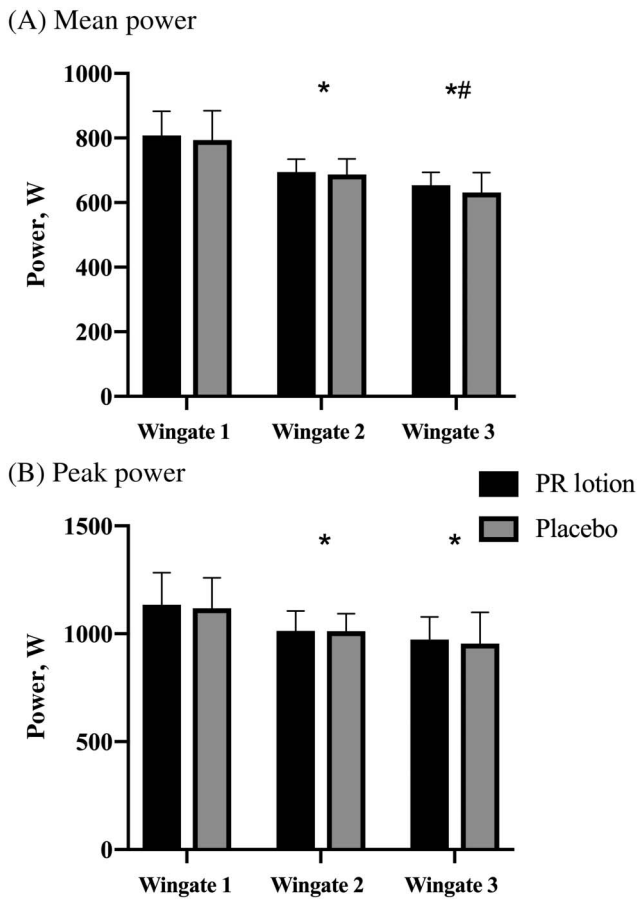


**Figure 5** — (A) Mean power and (B) peak power for the PR lotion and placebo conditions during study B. *Significant decrease compared with Wingate 1. #Significant decrease compared with Wingate 2.

methods. Specific to exercise performance, time to volitional exhaustion in recreationally active individuals (study A) was similar between PR lotion and oral $NaHCO_3$ supplementation, while no beneficial effect of PR lotion was observed during a fixed performance task in moderately trained athletes when compared with a placebo (study B).

Topical application of $NaHCO_3$ via PR lotion, even when matched for $NaHCO_3$ doses ingested orally, does not appear to alter systemic acid–base balance, as changes in blood buffering capacity were only observed after oral $NaHCO_3$ ingestion (6.5 [1.6] vs 1.5 [0.6] mmol/L increase in $HCO_3^-$ at 90 min postsupplementation). This finding was confirmed in study B, where no differences in acid–base status were evident between the PR lotion and placebo conditions when applied 30 minutes preexercise. As no peer-reviewed data exist on this product, and without the ability to trace $NaHCO_3$ across the multiple layers of skin, we matched the product gram for gram to a typical oral load of $NaHCO_3$ that has consistently demonstrated robust increases in systemic blood buffering capacity.[19–21] In the absence of any increased systemic changes in [$HCO_3^-$], our findings suggest that topical application of $NaHCO_3$ in the quantities administered in the present studies were not reflected in the wider circulation. Given recent evidence we cannot discount any localized effect of the menthol in the lotion,[22] however, any proposed mechanism of action for a potential performance benefit attributed to $NaHCO_3$ (which was not seen here) remains unclear.

The premise for a $NaHCO_3$ administration method that minimizes exposure to the stomach has a strong rationale. Recently, this idea was investigated in a postgastric bypass patient, where $NaHCO_3$ supplementation resulted in some of the highest [$HCO_3^-$] increases reported in the literature (>12 nM) with minimal side effects.[23] This finding was attributed to the decreased exposure time in the stomach, where large amounts of $HCO_3^-$ are usually lost in the neutralization of gastric acid.[11] In a more applied context, delayed release capsules have demonstrated the potential to increase pH and $HCO_3^-$



Authenticated t_lambrinides@yahoo.com | Downloaded 06/23/20 08:35 PM UTC

**Table 2 Blood pH, Bicarbonate, and Lactate Concentrations for Study B at Baseline, Preexercise, and Postexercise in the PR Lotion and Placebo Lotion Conditions, Mean (SD)**

| | PR lotion | | | Placebo | | |
|---|---|---|---|---|---|---|
| | Baseline | Preexercise | Postexercise | Baseline | Preexercise | Postexercise |
| pH | 7.42 (0.033) | 7.40* (0.038) | 7.13*,** (0.067) | 7.43 (0.046) | 7.40* (0.041) | 7.15*,** (0.083) |
| Bicarbonate, mmol/L | 25.44 (1.69) | 22.62* (2.64) | 10.66*,** (2.17) | 25.59 (1.89) | 23.36* (2.46) | 10.96*,** (2.20) |
| Lactate, mmol/L | 0.89 (0.33) | 2.21* (1.43) | 19.16*,** (1.95) | 1.47 (0.68) | 2.13 (0.76) | 18.88*,** (2.2) |

*Significantly different from baseline. **Significantly different from preexercise.

concentrations while minimizing GI symptoms, an affect attributed to the gastro-resistant properties of enteric-coated capsules.[24] It is thought that the polymer barrier of these capsules makes them stable at low pH and therefore resistant to the acidity in the stomach, ultimately allowing them to be broken down and absorbed in the more alkaline environment of the intestine.[23,24] Despite the potential benefits of bypassing the stomach, we saw no evidence of transdermal $NaHCO_3$ penetration from our topical administration of the PR lotion at either 30 minutes (study B) or 90 minutes (study A) postapplication. As such, it is likely that oral $NaHCO_3$ delivery methods remain the most efficacious method to increase blood buffering capacity; particularly, the oral $NaHCO_3$ supplementation protocol used in study A, as it did not heighten GI distress.

As anticipated, oral $NaHCO_3$ ingestion augmented blood $HCO_3^-$, pH, and lactate levels during rest (with the exception of lactate) and recovery in study A. The greater postexercise blood lactate concentrations are likely associated with the increased electrochemical gradient caused by the induced alkalosis, and the increased $HCO_3^-$ concentrations likely facilitated enhanced $H^+$ removal from the muscle.[25] Previous research has suggested that a minimum "alkalinity threshold" is needed (elevations in $HCO_3^-$ and pH of 6 mmol/L and 0.05 a.u., respectively) to improve the likelihood of inducing an ergogenic effect.[16,26] Furthermore, time to exhaustion tasks have demonstrated the largest ergogenic effect as a performance measure, compared with power or performance time.[5] Despite study A utilizing a quasi-time to exhaustion protocol and blood parameters meeting the suggested threshold ($HCO_3^-$: +6.5 mmol/L and pH: +0.07), oral $NaHCO_3$ ingestion did not appear to influence exercise performance. This finding must be taken in context; however, given the lack of a true control condition in study A. One could argue that the absence of a performance improvement in the oral $NaHCO_3$ condition does not establish the efficacy of PR lotion, but rather questions the efficacy of $NaHCO_3$ supplementation. Indeed, the lack of performance improvement in the oral $NaHCO_3$ condition is a limitation of this study and does ot support the large body of work suggesting that oral $NaHCO_3$ supplementation can positively impact performance.[4,5,15,26] However, it is likely the lack of an ergogenic effect in this instance may reflect the large interindividual and intraindividual variability in blood buffering kinetics associated with oral $NaHCO_3$ supplementation.[6,27] As such, these findings further highlight the need for supplementation protocols to be trialed before being implemented in a competition setting.[28]

It is also possible that the lack of performance improvement in study A after oral $NaHCO_3$ supplementation may be explained by the inclusion of a recreationally active population, as their inability to complete 2 maximal time-to-exhaustion efforts in a reliable manner may have contributed to the highly variable responses reported (Figure 3). This is supported by previous research suggesting that the caliber of athlete may influence the effectiveness of oral $NaHCO_3$ supplementation, with a larger ergogenic effect evident as athlete training status increases.[4] Therefore, study B was designed with the understanding that training status may have confounded any ergogenic benefit induced by $NaHCO_3$ supplementation in study A. This, coupled with the potential for a topical application to avoid complications experienced with oral ingestion (eg, GI discomfort), was a strong enough rationale to focus exclusively on performance outcomes in study B. Although we acknowledge the limitation of not including an oral $NaHCO_3$ load in study B, given the athletic population used in this study, we did not feel we would be able to adequately blind this population given the likelihood of our participants being familiar with even the faintest of side effects (eg, bloating, burping) experienced during an oral $NaHCO_3$ load. Regardless, application of the PR lotion 30 minutes prior to exercise, as in study A, did not alter blood buffering capacity nor did it enhance performance.

## Practical Applications

There is robust evidence in the literature to suggest that when carefully planned, oral $NaHCO_3$ supplementation between 0.2 and 0.4 g/kg BW can improve athlete performance by ~2% to 3%.[4,5] The current investigation failed to observe any changes in blood buffering capacity resulting from topical $NaHCO_3$ application in a concentration equivalent to 0.3 g/kg BW, and no ergogenic effect on performance was evident. Given these observations, we advise that oral ingestion rather than topical application of $NaHCO_3$ remains the preferred supplementation method for increasing blood buffering capacity and potentially improving performance tasks 1 to 7 minutes in duration.[4,5] Accordingly, and given the recognized GI issues that accompany the use of this supplement, athletes should aim to individualize their supplementation protocol in an attempt to maximize the ergogenic effect while minimizing GI discomfort. However, as with any dietary supplement, consultation with a sports nutrition professional is strongly advised prior to inclusion in an athlete's nutrition plan.[1,28]

## Conclusions

In both studies, we found no evidence for PR lotion to improve blood buffering capacity or enhance exercise performance, suggesting that $NaHCO_3$ has limited transdermal absorption capacity into the wider circulation. As such, a greater understanding of the proposed mechanisms by which $NaHCO_3$ is thought to cross the skin barrier is required before PR lotion can be considered as a viable alternative to oral $NaHCO_3$ ingestion.

### Acknowledgments

The authors wish to thank all those who participated in this study. Amp Human provided the lotions used in this study free of charge but had no



Authenticated t_lambrinides@yahoo.com | Downloaded 06/23/20 08:35 PM UTC

influence over the study design or in the preparation of the manuscript. The results of this study do not constitute endorsement of the product by the authors or the journal. The authors declare no potential conflict of interest at the time the study was conducted; however, during the peer-review phase of this manuscript, the senior author (J.S.) accepted a role on Amp Human's scientific advisory board.

## References

1. Maughan RJ, Burke LM, Dvorak J, et al. IOC consensus statement: dietary supplements and the high-performance athlete. *Int J Sport Nutr Exerc Metab*. 2018;28(2):104–125. doi:10.1123/ijsnem.2018-0020
2. Lancha Junior AH, Painelli Vde S, Saunders B, Artioli GG. Nutritional strategies to modulate intracellular and extracellular buffering capacity during high-intensity exercise. *Sports Med*. 2015;45(1): 71–81. doi:10.1007/s40279-015-0397-5
3. Requena B, Zabala M, Padial P, Feriche B. Sodium bicarbonate and sodium citrate: ergogenic aids? *J Strength Cond Res*. 2005;19(1): 213–224. PubMed ID: 15705037
4. Carr AJ, Hopkins WG, Gore CJ. Effects of acute alkalosis and acidosis on performance. *Sports Med*. 2011;41(10):801–814. PubMed ID: 21923200 doi:10.2165/11591440-000000000-00000
5. Peart DJ, Siegler JC, Vince RV. Practical recommendations for coaches and athletes: a meta-analysis of sodium bicarbonate use for athletic performance. *J Strength Cond Res*. 2012;26(7):1975–1983. PubMed ID: 22505127 doi:10.1519/JSC.0b013e3182576f3d
6. Jones RL, Stellingwerff T, Artioli GG, Saunders B, Cooper S, Sale C. Dose–response of sodium bicarbonate ingestion highlights individuality in time course of blood analyte responses. *Int J Sport Nutr Exerc Metab*. 2016;26(5):445–453. PubMed ID: 27098290 doi:10.1123/ijsnem.2015-0286
7. Siegler JC, Marshall PW, Bray J, Towlson C. Sodium bicarbonate supplementation and ingestion timing: does it matter? *J Strength Cond Res*. 2012;26(7):1953–1958. PubMed ID: 21964428 doi:10.1519/JSC.0b013e3182392960
8. McNaughton LR. Bicarbonate ingestion: effects of dosage on 60 s cycle ergometry. *J Sports Sci*. 1992;10(5):415–423. PubMed ID: 1331493 doi:10.1080/02640419208729940
9. Burke LM, Pyne DB. Bicarbonate loading to enhance training and competitive performance. *Int J Sports Physiol Perform*. 2007;2(1): 93–97. PubMed ID: 19255457 doi:10.1123/ijspp.2.1.93
10. Carr AJ, Gore CJ, Dawson B. Induced alkalosis and caffeine supplementation: effects on 2,000-m rowing performance. *Int J Sport Nutr Exerc Metab*. 2011;21(5):357–364. PubMed ID: 21799214 doi:10.1123/ijsnem.21.5.357
11. Turnberg LA, Fordtran JS, Carter N, Rector FC. Mechanism of bicarbonate absorption and its relationship to sodium transport in the human jejunum. *J Clin Invest*. 1970;49(3):548–556. PubMed ID: 5415681 doi:10.1172/JCI106265
12. Kahle LE, Kelly PV, Eliot KA, Weiss EP. Acute sodium bicarbonate loading has negligible effects on resting and exercise blood pressure but causes gastrointestinal distress. *Nutr Res*. 2013;33(6):479–486. PubMed ID: 23746564 doi:10.1016/j.nutres.2013.04.009
13. Cameron SL, McLay-Cooke RT, Brown RC, Gray AR, Fairbairn KA. Increased blood pH but not performance with sodium bicarbonate supplementation in elite rugby union players. *Int J Sport Nutr Exerc Metab*. 2010;20(4):307–321. PubMed ID: 20739719 doi:10.1123/ijsnem.20.4.307
14. Burke LM. Practical considerations for bicarbonate loading and sports performance. *Nestle Nutr Inst Workshop Ser*. 2013;75: 15–26. PubMed ID: 23765347 doi:10.1159/000345814
15. Driller MW, Gregory JR, Williams AD, Fell JW. The effects of serial and acute $NaHCO_3$ loading in well-trained cyclists. *J Strength Cond Res*. 2012;26(10):2791–2797. PubMed ID: 23001395 doi:10.1519/JSC.0b013e318241e18a
16. Carr AJ, Slater GJ, Gore CJ, Dawson B, Burke LM. Effect of sodium bicarbonate on [$HCO_3^-$], pH, and gastrointestinal symptoms. *Int J Sport Nutr Exerc Metab*. 2011;21(3):189–194. PubMed ID: 21719899 doi:10.1123/ijsnem.21.3.189
17. Currell K, Jentjens RL, Jeukendrup AE. Reliability of a cycling time trial in a glycogen-depleted state. *Eur J Appl Physiol*. 2006; 98(6):583–589. PubMed ID: 17016701 doi:10.1007/s00421-006-0305-7
18. Borg GA. Psychophysical bases of perceived exertion. *Med Sci Sports Exerc*. 1982;14(5):377–381. PubMed ID: 7154893 doi:10.1249/00005768-198205000-00012
19. Siegler JC, Midgley AW, Polman RC, Lever R. Effects of various sodium bicarbonate loading protocols on the time-dependent extracellular buffering profile. *J Strength Cond Res*. 2010;24(9):2551–2557. PubMed ID: 20040895 doi:10.1519/JSC.0b013e3181aeb154
20. Price MJ, Singh M. Time course of blood bicarbonate and pH three hours after sodium bicarbonate ingestion. *Int J Sports Physiol Perform*. 2008;3(2):240–242. PubMed ID: 19208932 doi:10.1123/ijspp.3.2.240
21. Renfree A. The time course for changes in plasma [$H^+$] after sodium bicarbonate ingestion. *Int J Sports Physiol Perform*. 2007;2(3): 323–326. PubMed ID: 19168932 doi:10.1123/ijspp.2.3.323
22. Gillis DJ, Vellante A, Gallo JA, D'Amico AP. Influence of menthol on recovery from exercise-induced muscle damage. *J Strength Cond Res*. 2020;34(2):451–462. PubMed ID: 30161086 doi:10.1519/JSC.0000000000002833
23. de Oliveira LF, Saunders B, Artioli GG. Is bypassing the stomach a means to optimize sodium bicarbonate supplementation? A case study with a postbariatric surgery individual. *Int J Sport Nutr Exerc Metab*. 2018;28(6):660–663. PubMed ID: 29722576 doi:10.1123/ijsnem.2017-0394
24. Hilton NP, Leach NK, Sparks SA, et al. A novel ingestion strategy for sodium bicarbonate supplementation in a delayed-release form: a randomised crossover study in trained males. *Sports Med Open*. 2019;5(1):4. PubMed ID: 30680463 doi:10.1186/s40798-019-0177-0
25. Siegler JC, Keatley S, Midgley AW, Nevill AM, McNaughton LR. Pre-exercise alkalosis and acid-base recovery. *Int J Sports Med*. 2008;29(7):545–551. PubMed ID: 18004683 doi:10.1055/s-2007-989261
26. Bishop D, Claudius B. Effects of induced metabolic alkalosis on prolonged intermittent-sprint performance. *Med Sci Sports Exerc*. 2005;37(5):759–767. PubMed ID: 15870629 doi:10.1249/01.MSS.0000161803.44656.3C
27. Saunders B, Sale C, Harris RC, Sunderland C. Sodium bicarbonate and high-intensity-cycling capacity: variability in responses. *Int J Sports Physiol Perform*. 2014;9(4):627–632. PubMed ID: 24155093 doi:10.1123/ijspp.2013-0295
28. Peeling P, Binnie MJ, Goods PS, Sim M, Burke LM. Evidence-based supplements for the enhancement of athletic performance. *Int J Sport Nutr Exerc Metab*. 2018;28(2):178–187. PubMed ID: 29465269 doi:10.1123/ijsnem.2017-0343



Authenticated t_lambrinides@yahoo.com | Downloaded 06/23/20 08:35 PM UTC